

Total No. of Questions : 9]

PA-4294

SEAT No. :

[Total No. of Pages : 4

[5924]-3

F.E.

ENGINEERING CHEMISTRY
(2019 Pattern) (Semester - I) (107009)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates :

- 1) Question No. 1 is compulsory. Solve Q.No.2 or Q.No.3, Q.No.4 or Q.No.5, Q.No.6 or Q.No.7, Q.No.8 or Q.No.9.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Use of logarithmic tables slide rule, Mollier charts, electronic pocket calculator and steam tables is allowed.
- 5) Assume suitable data, if necessary.

Q1) Multiple Choice Questions :

- a) Applications of quantum dots are [2]
 - i) Light emitting diode
 - ii) Photo detectors
 - iii) Biological analysis
 - iv) All of these
- b) CDS and DVDS can be made by using [1]
 - i) Polycarbonate
 - ii) Polyphenylene vinylene
 - iii) Kevlar
 - iv) Polyisoprene
- c) $NCV = GCV - \frac{H}{8} \times 587 \text{ Kcal/kg.}$ [1]
 - i) 0.9
 - ii) 9
 - iii) 0.09
 - iv) 90
- d) Kjeldhal's method is used for determination of [1]
 - i) % H
 - ii) % N
 - iii) % S
 - iv) % C

P.T.O.

- e) Electromagnetic radiations consist of _____ oscillating perpendicular to each other. [2]
- i) Electric and magnetic field ii) Electric and magnetic radiation
iii) Charged particle iv) All of these

- f) Infrared spectroscopy is also called as _____. [1]
- i) Rotational spectroscopy
ii) Electronic excitation spectroscopy
iii) Vibrational spectroscopy
iv) Nuclear excitation spectroscopy

- g) Which of the following film is protective? [1]
- i) Porous ii) Non porous
iii) Volatile iv) None of these

- h) Tinning is the example of [1]
- i) Anodic coating ii) Cathodic coating
iii) Neither anode nor cathode iv) Both anode and cathode

Q2) a) Define conducting polymers. What is the structural requirement for a polymer to show conductive property? Give doping reactions two properties and two applications of conducting polymer. [6]

b) Explain the structure of graphene with the help of diagram and mention its two properties and two applications. [5]

c) What are biodegradable polymers? Explain the three factors responsible for biodegradation. [4]

OR

Q3) a) Define Polymer composites. Discuss the classification on the basis of reinforcement with any two applications. [6]

b) Discuss the different types of carbon nanotubes with respect to their structure. Give any two application of carbon nanotubes. [5]

c) What are the nanomaterials? Classify any two on the basis of dimensions. Give any two applications. [4]

Q4) a) Explain in brief the process with diagram for distillation of crude petroleum. Give composition, boiling range and uses of two fractions obtained. [6]

b) What is Power alcohol? Give its preparation with reactions and any two advantages. [5]

c) Calculate the NCV of fuel containing 6% hydrogen with the following data of Bomb calorimeter experiment. [4]

i) Mass of fuel = 0.850 gm

ii) Mass of water in calorimeter = 2200 gm

iii) Water equivalent of calorimeter set = 750 gm

iv) Rise in temperature = 2.82°C

OR

Q5) a) What is proximate analysis? Mention the principle involved in the analysis of each of these constituents. Give the significance of any one constituent. [6]

b) Explain the production of hydrogen by steam reforming of methane and coke with reaction conditions. [5]

c) 0.30 gm of coal sample on burning in combustion chamber in current of pure O_2 was found to increase weight of CaCl_2 U tube by 0.11 gm and KOH U tube by 0.62 gm. Find carbon hydrogen % in the coal sample. [4]

Q6) a) Draw the block diagram of single beam UV visible spectrophotometer. Explain its four components and give their function. [6]

b) Give the principle of IR spectroscopy. Explain the fundamental modes of stretching vibrations. [5]

c) Define the following terms : [4]

i) Auxochrome

ii) Chromophore

iii) Bathochromic shift

iv) Hyperchromic shift

OR

Q7) a) Explain different types of electronic transitions that occurs in an organic molecule after absorbing UV radiations. [6]

b) What are the conditions of absorption of IR radiations by the molecule? Explain any two applications of IR spectroscopy. [5]

c) Calculate the fundamental modes of vibrations in the following molecules. [4]

i) H_2O

ii) NH_3

iii) CH_4

iv) NO

Q8) a) Explain the mechanism of dry corrosion. Give the nature of oxide film and reaction in oxidation corrosion of Mg and Cr. [6]

b) What is the principle of Cathodic protection? Explain sacrificial anodic protection of metal. [5]

c) Explain any four factors affecting the rate of corrosion related to metal. [4]

OR

Q9) a) Define electroplating. Explain the process with diagram and reactions. Give any two applications of electroplating. [6]

b) Explain the mechanism of wet corrosion by Hydrogen evolution with diagram and reactions. [5]

c) What is Galvanising? Explain the process with neat labelled diagram to protect iron from corrosion. [4]

